

Projections: The Geometry of Least Squares

Least squares chooses the point of a subspace that is closest to the data. Everything else in this chapter follows from that one statement: the normal equations, why fitted values are unique when coefficients are not, the orthogonal decompositions behind every sum of squares, the Frisch–Waugh–Lovell theorem, R^2 as a squared cosine, leverage, generalized least squares, and how the projection should actually be computed.

What you need. [Chapter 1](#) (rank, inverses, eigenvalues), [Chapter 5](#) (the linear model and least squares). Random vectors ([Chapter 2](#)) appear only in a few remarks.

6.-1.1 Roadmap

- [Least squares as a nearest-point problem](#)
- [The subspaces we need](#)
- [Orthogonal projection](#)
- [Least squares via projection](#)
- [Nested models and sequential projection](#)
- [The Frisch–Waugh–Lovell theorem](#)
- [Reparameterization, centring and \$R^2\$](#)
- [Leverage as geometry](#)
- [Other inner products](#)
- [Computation](#)
- [Projection in the population: best linear prediction](#)
- [Summary and notes](#)